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Supplementary information

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Engineered circular RNA compatible with complete nucleoside modification and rolling circle translation through a Cap-independent translation enhancer

In the format provided by the authors and unedited

Table of Contents

Supplementary Figure

Supplementary Fig. 1 | The gating strategy of flow cytometry analysis.

Supplementary Tables

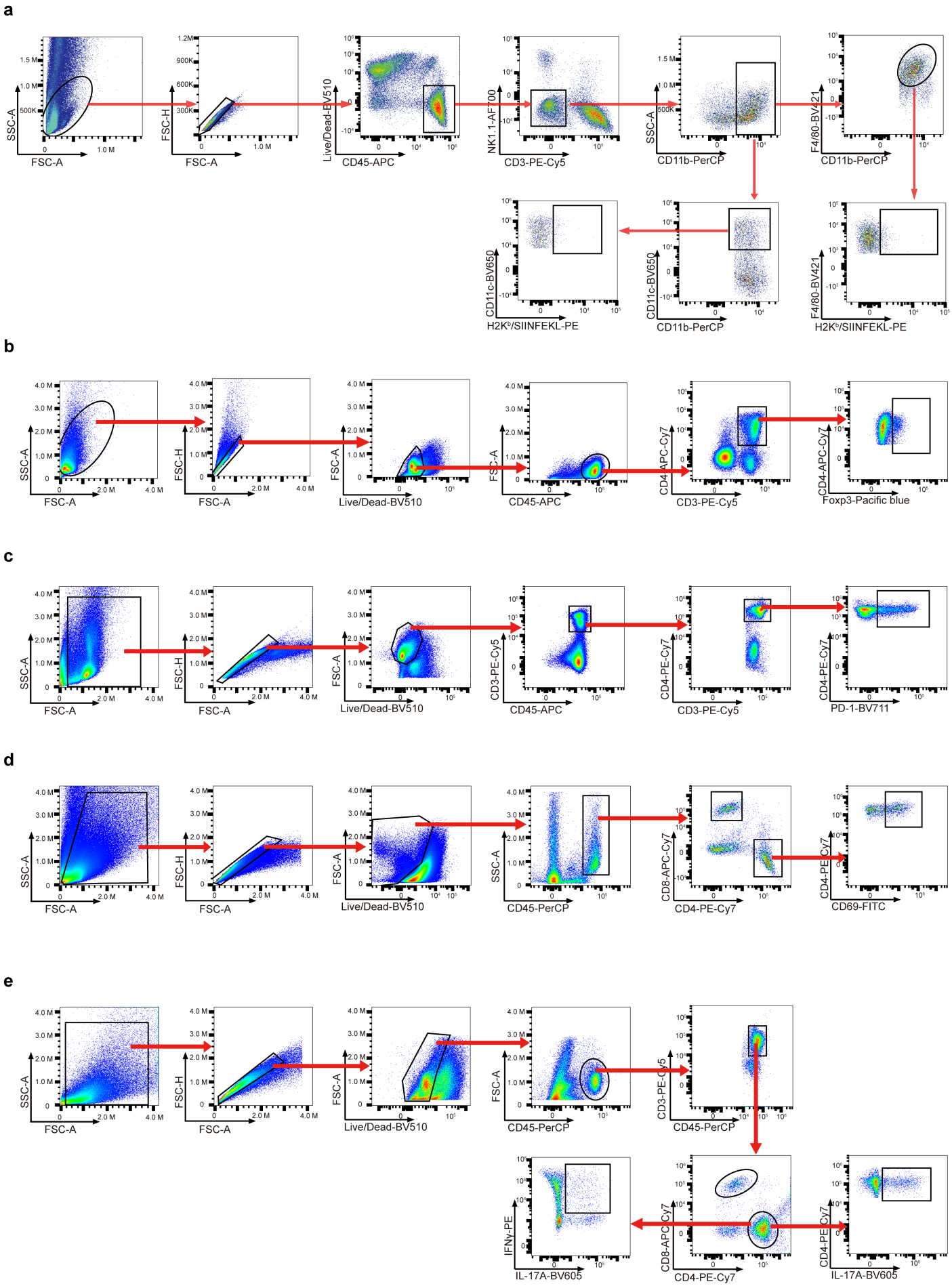
Supplementary Table 1 | IRES sequences.

Supplementary Table 2 | CITE and H2SL sequences.

Supplementary Table 3 | Typical RNA and DNA sequences information.

Supplementary Table 4 | Primer sequences.

Supplementary Table 5 | List of antibodies used in this study.


a
SSC-A
FSC-A
FSC-H
FSC-A
Live/Dead-BV510
CD45-APC
NK1.1-AF700
CD3-PE-Cy5
SSC-A
CD11b-PerCP
F4/80-BV421
CD11b-PerCP
CD11c-BV650
H2Kᵇ/SIINFEKL-PE
CD11c-BV650
CD11b-PerCP
F4/80-BV421
H2Kᵇ/SIINFEKL-PE
b
SSC-A
FSC-A
FSC-H
FSC-A
FSC-A
Live/Dead-BV510
FSC-A
CD45-APC
CD4-APC-Cy7
CD3-PE-Cy5
CD4-APC-Cy7
Foxp3-Pacific blue
c
SSC-A
FSC-A
FSC-H
FSC-A
FSC-A
Live/Dead-BV510
CD3-PE-Cy5
CD45-APC
CD4-PE-Cy7
CD3-PE-Cy5
CD4-PE-Cy7
PD-1-BV711
d
SSC-A
FSC-A
FSC-H
FSC-A
FSC-A
Live/Dead-BV510
SSC-A
CD45-PerCP
CD8-APC-Cy7
CD4-PE-Cy7
CD4-PE-Cy7
CD69-FITC
e
SSC-A
FSC-A
FSC-H
FSC-A
FSC-A
Live/Dead-BV510
FSC-A
CD45-PerCP
CD3-PE-Cy5
CD45-PerCP
IFNγ-PE
IL-17A-BV605
CD8-APC-Cy7
CD4-PE-Cy7
CD4-PE-Cy7
IL-17A-BV605

Supplementary Fig. 1 | The gating strategy of flow cytometry analysis. a, Gating strategy of flow cytometry analysis for the infiltrating immune cells in the tumor microenvironment. **b**, Gating strategy of flow cytometry analysis for Treg. **c**, Gating strategy of flow cytometry analysis for PD-1$^+$ CD4$^+$ T cells. **d**, Gating strategy of flow cytometry analysis for CD69$^+$ CD4$^+$ T cells. **e**, Gating strategy of flow cytometry analysis for the population of CD4$^+$ T cell subsets producing pro-inflammatory cytokines (IFN-γ and IL-17A).

Supplementary Table 1 | IRES sequences.

IRES name	Sequence (5'-3')
EchoV-E11	UUAAAACAGCCUGUGGGUUGUACCCACCCACAGGGCCCACUGGGCGCUAGCACAC UGGUAUCACGGUACCUUUGUGCGCCUGUUUUAUACCCCCUUCCCGCAACCGCAAA UUUAGAAGCAAAGCUAACCCGAUCGAUAGCGGAUGCGCAUGCCAGCCGCAUUUUG AUCAAGUACUUCUGUUUCCCCGGACCGAGUAUCAAUAGACUGCUCACGCGGUUGA AGGAGAAAACGUCCGUUACCCGACCAACUACUUCGAGAAACCUAGUAACAUCAUGA AUGUUGCAGGGCGUUUCGAUCAGCACGACCCUGGUGUAGAUCAGGCUGAUGAGUC ACCGCAUUCCCCACGGGUGACCGUGGCGGUGGCUGCGUUGGCGGCCUGCCUAUG GGGUGACCCAUAGGACGCUCUAAUACGGACAUGGUGCGAAGAGUCUAUUGAGCUA GUUGGUAGUCCUCCGGCCCCUGAAUGCGGUUAAUCCUAACUGCGGACGACAUACC CCUAAUCCAAGGGGCAGUGUGUCGUAACGGGCAACUCUGCAGCGGAACCGACUAC UUUGGGUGUCCGUGUUUCCUUUUAUUUUUAUACUGGCUGCUUAUGGUGACAAUCU CAGAGUUGUUACCAUAUAGCUAUUGGAUUGGCCAUCCGGUGAGCAACAGAGCUGU CAUUUAUCAGUUUGUUGGCUUUAUACCUCUAAAUCACACGGUUUUUUUUUUUUGG AACGCUUGUAUUCAUCUUAACCCUCAAUAAGGCAAA
CVA20	UUAAAACAGCUCUGGGGUUGUACCCACCCCAGAGGCCCACGUGGCGGCUAGUACU CCGGUAUUACGGUACCCUUGUACGCCUGUUUUAUACUCCCUCCCCUCGUAACUUAG AAGCACAAAACCAAGUUCAAUAGAAGGGGGUGCAAACCAGUACCACCACGAACAAG CAUUUCUGUUUCCCCGGUGAUGUUGUAUAGACUGCCCGCGCGGUUGAAAGCAACG GUACCGUUACCCGCUCAAGUACUUCGAGAAGCCUAGUAUUACCUUGGAAUCUUCG AUGCGUUGCGUUCAGCACUCGACCCUGGAGUGUAGCUUAGGCUGAUGAGUCUGG ACGUCCCUCACCGGUGACGGUGGUCCAGGCUGCGUUGGCGGCCUACCUAUGGCU AACGCCAUAGGACGCUAGUUGUGAACAAGGUGUGAAGAGCCUAUUGAGCUACUUG AGAGUCCUCCGGCCCCUGAAUGCGGCUAAUCCUAACCAUGGAGCAGGCGGUCACA GACCAGUGACUAGCUUGUCGUAAUGCGCAAGUCUAUGGCGGAACCGACUACUUUG GGUGUCCGUGUUUCCUUUUAUUUUUAUUAUGGCUGCUUAUGGUGACAAUCAUUGA UUGUUAUCAUAAAGCGAAUUGGAUUGGCCAUCCGGUGAAAGCGAGACUUACUAUU UACUUACUUGUUGGACUUACCACACUUAAUACAUUUAUUCUAGGUGUCACUUGUAU AGCAAUUAGAAUCAAACAGUUGCAUCAUA
Aichi Virus	UUUGAAAAGGGGGUGGGGGGGCCUCGGCCCCCUCACCCUCUUUUCCGGUGGUCU GGUCCCGGACCACCGUUACUCCAUUCAGCUUCUUCGGAACCUGUUCGGAGGAAUU AAACGGGCACCCAUACUCCCCCCACCCCCCUUUUGUAACUAAGUAUGUGUGCUCG UGAUCUUGACUCCCACGGAACGGACCGAUCCGUUGGUGAACAAACAGCUAGGUCC ACAUCCUCCCUUCCCCCUGGGAGGGCCCCCGCCCUCCCACAUCCUCCCCCCAGCCU GACGUAUCACAGGCUGUGUGAAGCCCCCGCGAAAGCUGCUCACGUGGCAAUUGUG GGUCCCCCCUUCAUCAAGACACCAGGUCUUUCCUCCUUAAGGCUAGCCCCGGCGU GUGAAUUCACGUUGGGCAACUAGUGGUGUCACUGUGCGCUCCCAAUCUCGGCCGC GGAGUGCUGUUCCCCAAGCCAAACCCCUGGCCCUUCACUAUGUGCCUGGCAAGCA UAUCUGAGAAGGUGUUCCGCUGUGGCUGCCAACCUGGUGACAGGUGCCCCAGUG UGCGUAACCUUCUUCCGUCUCCGGACGGUAGUGAUUGGUUAAGAUUUGGUGUAAG GUUCAUGUGCCAACGCCCUGUGCGGGAUGAAACCUCUACUGCCCUAGGAAUGCCA GGCAGGUACCCCACCUCCGGGUGGGAUCUGAGCCUGGGCUAAUUGUCUACGGGU AGUUUCAUUUCCAAUCCUUUUAUGUCGGAGUC
SwineVesicular	UUAAAACAGCCUGUGGGUUGUUCCCACCCACAGGGCCCACUGGGCGCUAGCACAC UGGUAUCACGGUACCUUUGUGCGCCUGUUUGACUUACCCUCCCCCAAACGCAACUU AGAAGCACAACUUAAAUGGUCAAUAGGCGGCUCAGUAUGCCAACUGAGUCUCGAU CAAGCACUUCUGUUACCCCGGACUGAGUACCAAUAGGCUGCUCACGCGGCUGAAG GGGAAACCGUUCGUUACCCGACUAACUACUUCGAGAAACCUAGUACCACCAUGAAA GUUGCGCAGCGUUUCGCUCCGCACAACCCCAGUGUAGAUCAGGUCGAUGAGUCAC CGCAAACCCCACGGGCGACCGUGGCGGUGGCUGCGCUGGCGGCCUGCCCAUGGG GCAACUCAUGGGACGCUUCAAUACUGACAUGGUGCGAAGAGUCUAUUGAGCUAGU UGGUAGUCCUCCGGCCCCUGAAUGCGGCUAAUCCUAACUGCGGAGCAGAUACCCA CGCACCAGUGGGCAGUCUGUCGUAAUGGGCAACUCUGCAGCGGAACCGACUACUU UGGGUGUCCGUGUUUCCUUUUGUUCUUAUACUGGCUACUUAUGGUGACAAUUGAG AGAUUGUAACCAUAUUGCUAUUGGAUUGGCCACCCGGCGACGAAUAGAACAGUUG

	CUUACCUGUUUGUUAGUCUCGUAUCACUGAACUACAAAUCCUUAAACACCCUUUAA UUUCGUCAUAACGCUCAAUACGUUAAA
EV107	UUAAAACAGCCUGUGGGUUGUUCCCACCCGCAGGGCCCACUGGGCGCUAGCACAC UGGUAUCCCGGUACCCUUGUGCGCCUGUUUUAUAUACCCUCCCCCUUAUGUAACU UAGAAGUAUGAUUCAAACGGUCGACAGGCGGCUCAGUGCACCAACUGAGUCAUGA CCAAGCACUUCUGUUACCCCGGACUGAGUAUCAAUAAGCUGUUCACACGGCUGAA GGAGAAAACGUUCGUUACCCGGCCAAUUACUUCGAGAAACCUAGUACCACCAUGAA GGUUGCGCAGUGUUUCGCUCCACACAACCCCAGUGUAGAUCAGGUCGAUGAGUCA CCGCAUUCCCCACGGGCGACCGUGGCGGUGGCUGCGUUGGCGGCCUGCCCAUGG GGCAACCCAUGGGACGCUUCAAUACUGACAUGGUGUGAAGAGUCUAUUGAGCUAA UUGGUAGUCCUCCGGCCCCUGAAUGCGGCUAAUCCCAACUGUGGAGCAGAUACUC ACAAACCAGUGAGCGGUCUGUCGUAACGGGCAACUCCGCAGCGGAACCGACUACU UUGGGUGUCCGUGUUUCUUUUUAUUCUUACAUUGGCUGCUUAUGGUGACAAUUGA CAAAUUGUUACCAUAUAGCUAUUGGAUUGGCCAUCCGGUGACAAACAGAGCUAUU GUUUACUUGUUUGUUGGUUUCAUACCAUUAAAUUACAAGGUCUUAGAAACUCUCAA CUUUAUUUUGACACUCAAUACAGCAAA
CVA3	UUAAAACAGCCUGUGGGUUGUACCCACCCACAGGGCCCACUGGGCGCUAGCACAC UGGUAUUACGGUACCUUUGUGCGCCUGUUUUAUACCCCCCCCAACCUCGAAACUU AGAAGUAAAGCAAACCCGAUCAAUAGCAGGUGCGGCGCACCAGUCGCAUCUUGAU CAAGCACUUCUGUAACCCCGGACCGAGUAUCAAUAGACUGCUCACGCGGUUGAAG GAGAAAACGUUCGUUACCCGGCUAACUACUUCGAGAAACCCAGUAGCAUCAUGAAA GUUGCAGAGUGUUUCGCUCAGCACUACCCCCGUGUAGAUCAGGCCGAUGAGUCAC CGCACUUCCCCACGGGCGACCGUGGCGGUGGCUGCGUUGGCGGCCUGCCUAUGG GGCAACCCAUAGGACGCUCUAAUACGGACAUGGUGCGAAGAGUCUAUUGAGCUAG UUAGUAGUCCUCCGGCCCCUGAAUGCGGCUAAUCCUAACUGCGGAGCACAUACCC UUAAUCCAAAGGGCAGUGUGUCGUAACGGGUAACUCUGCAGCGGAACCGACUACU UUGGGUGUCCGUGUUUCCUUUUAAUUUUUACUGGCUGCUUAUGGUGACAAUUGAG GAAUUGUUGCCAUAUAGCUAUUGGAUUGGCCAUCCGGUGACUAACAGAGCUAUUG UGUUCCAAUUUGUUGGAUUUACCCCGCUCACACUCACAGUCGUAAGAACCCUUCA UUACGUGUUAUUUCUCAACUCAAGAAA
PV1	UUAAAACAGCUCUGGGGUUGUACCCACCCCAGAGGCCCACGUGGCGGCUAGUACU CCGGUAUUGCGGUACCCUUGUACGCCUGUUUUAUACUCCCUUCCCGUAACUUAGA CGCACAAAACCAAGUUCAAUAGAAGGGGGUACAAACCAGUACCACCACGAACAAGC ACUUCUGUUUCCCCGGUGAUGUCGUAUAGACUGCUUGCGUGGUUGAAAGCGACG GAUCCGUUAUCCGCUUAUGUACUUCGAGAAGCCCAGUACCACCUCGGAAUCUUCG AUGCGUUGCGCUCAGCACUCAACCCCAGAGUGUAGCUUAGGCUGAUGAGUCUGGA CAUCCCUCACCGGUGACGGUGGUCCAGGCUGCGUUGGCGGCCUACCUAUGGCUA ACGCCAUGGGACGCUAGUUGUGAACAAGGUGUGAAGAGCCUAUUGAGCUACAUAA GAAUCCUCCGGCCCCUGAAUGCGGCUAAUCCCAACCUCGGAGCAGGUGGUCACAA ACCAGUGAUUGGCCUGUCGUAACGCGCAAGUCCGUGGCGGAACCGACUACUUUGG GUGUCCGUGUUUCCUUUUAUUUUAUUGUGGCUGCUUAUGGUGACAAUCACAGAUU GUUAUCAUAAAGCGAAUUGGAUUGGCCAUCCGGUGAAAGUGAGACUCAUUAUCUA UCUGUUUGCUGGAUCCGCUCCAUUGAGUGUGUUUACUCUAAGUACAAUUUCAACA GUUAUUUCAAUCAGACAAUUGUAUCACC
CVB5	UUAAAACAGCCUGUGGGUUGUUCCCACCCGCAGGGCCCACUGGGCGCCAGCACUC UGGUAUCACGGUACCUUUGUGCGCCUGUUUUAAAACCCUCUCCCCCAAUUUGAAAC UUAGAAGCAAUACACCUCGAUCAAUAGUAGGCAUGACACGCCAGCCAUGUCUUGAU CAAGCACUUCUGUUUCCCCGGACUGAGUAUCAAUAAACUGCUUGCGCGGUCGAAG GAGAAAACGUCCGUUACCCGACUAACUACUUCGAGAAACCCAGUAACACCAUGGAA AUUGCGGAGUGUUUCACUCAGCACAUUCCCAGUGUAGAUCAGGUCGAUGAGUCAC CGCAUUCCCCACGGGUGACCGUGGCGGUGGCUGCGCUGGCGGCCUGCCCAUGGG GCAACCCAUGGGACGCUUCAAUAUGGACAUGGUGUGAAGAGUCUAUUGAGCUAGU UAGUAGUCCUCCGGCCCCUGAAUGCGGCUAAUCCUAACUGCGGAGCACGUGCCUC CAUUCCAGGGGGUGGCGUGUCGUAACGGGCAACUCUGCAGCGGAACCGACUACUU UGGGUGUCCGUGUUUCUUUUAAUUUUAUACUGGCUGCUUAUGGUGACAAUUGAAA GAUUGUUGCCAUAUAGCUAUUGGAUUGGCCAUCCGGUAUCCAACAGAGCAAUUGU GUACCUUUUUGUUGGAUUUGUACCACUUACCAGAACAAGUUUUCAUACACUGUGU UACAUUAUUAGACUAAACACAGAAAA

CVB1	UUAAAACAGCCUGUGGGUUGUUCCCACCCACAGGGCCCAUUGGGCGCUAGCACUC UGGUAUCACGGUACCUUUGUGCGCCUGUUUUACAUCCCCUCCCCAAAUUGUAAUU UAGAAGUCUCACACACCGAUCAAUAGCAAGCGUGGCACACCAGCCAUGUUUUGAUC AAGCACUUCUGUUACCCGGACUGAGUAUCAAUAGACCGCUAACGCGGUUGAAGGA GAAAACGUUCGUUACCCGGCCAACUACUUCGAAAAACCUAGUAACACCAUGGAAGU UGCGGAGUGUUUCGCUCAGCACUACCCCAGUGUAGAUCAGGUCGAUGAGUCACCG CGUUCCCCACGGGCGACCGUGGCGGUGGCUGCGUUGGCGGCCUGCCUACGGGGA AACCCGUAGGACGCUCUAAUACAGACAUGGUGCGAAGAGUCUAUUGAGCUAGUUG GUAAUCCUCCGGCCCCUGAAUGCGGCUAAUCCUAACUGCGGAGCACAUACCCUCA AACCAGGGGGCAGUGUGUCGUAACGGGCAACUCUGCAGCGGAACCGACUACUUUG GGUGUCCGUGUUUCAUUUUAUUCCUAUACUGGCUGCUUAUGGUGACAAUUGACGG GUUGUUACCAUAUAGUUAUUGGAUUGGCCAUCCGGUGACUAACAGAGCAAUUAUA UAUCUCUUUGUUGGGUUUAUACCACUUAGCUUGAAAGAGGUUAAAACACUACAUCU CAUCAUUAAACUAAAUACAACAAA
CVB3	UUAAAACAGCCUGUGGGUUGAUCCCACCCACAGGCCCAUUGGGCGCUAGCACUCU GGUAUCACGGUACCUUUGUGCGCCUGUUUUAUACCCCCUCCCCCAACUGUAACUU AGAAGUAACACACACCGAUCAACAGUCAGCGUGGCACACCAGCCACGUUUUGAUCA AGCACUUCUGUUACCCCGGACUGAGUAUCAAUAGACUGCUCACGCGGUUGAAGGA GAAAGCGUUCGUUAUCCGGCCAACUACUUCGAAAAACCUAGUAACACCGUGGAAGU UGCAGAGUGUUUCGCUCAGCACUACCCCAGUGUAGAUCAGGUCGAUGAGUCACCG CAUUCCCCACGGGCGACCGUGGCGGUGGCUGCGUUGGCGGCCUGCCCAUGGGGA AACCCAUGGGACGCUCUAAUACAGACAUGGUGCGAAGAGUCUAUUGAGCUAGUUG GUAGUCCUCCGGCCCCUGAAUGCGGCUAAUCCUAACUGCGGAGCACACACCCUCA AGCCAGAGGGCAGUGUGUCGUAACGGGCAACUCUGCAGCGGAACCGACUACUUUG GGUGUCCGUGUUUCAUUUUAUUCCUAUACUGGCUGCUUAUGGUGACAAUUGAGAG AUCGUUACCAUAUAGCUAUUGGAUUGGCCAUCCGGUGACUAAUAGAGCUAUUAUA UAUCCCUUUGUUGGGUUUAUACCACUUAGCUUGAAAGAGGUUAAAACAUUACAAUU CAUUGUUAAGUUGAAUACAGCAAA
Enterovirus J	UUAAAAUAGCCUCAGGGUUGUUCCCACCCUGAGGGCCCACGUGGUGUAGUACUCU GGUAUUACGGUACCUUUGUACGCCUAUUUUAUACCCCCUUCCCCCAAGUAAUUUAG AAGCAAGCACAAACCAGUUCAGUAGUAAGCAGUACAAUCCAGUACUGUAAUGAACA AGUACUUCUGUUACCCCGGAAGGGUCUAUCGGUAAGCUGUACCCACGGCUGAAGA AUGACCUACCGUUAACCGGCUACCUACUUCGAGAAGCCUAGUAAUGCCGUUGAAG UUUUAUUGACGUUACGCUCAGCACACUACCCCGUGUGUAGUUUUGGCUGAUGAGU CACGGCACUCCCCACGGGCGACCGUGGCCGUGGCUGCGUUGGCGGCCAACCAAG GAGUGCAAGCUCCUUGGACGUCAUAUUACAGACAUGGUGUGAAGAGCCUAUUGAG CUAGGUGGUAGUCCUCCGGCCCCUGAAUGCGGCUAAUCCUAACUCCGGAGCAUAU CGGUGCGAACCAGCACUUGGUGUGUUGUAAUACGUAAGUCUGGAGCGGAACCGAC UACUUUGGGUGUCCGUGUUUCCUGUUUUAACUUUUAUGGCUGCUUAUGGUGACAA UUUAACAUUGUUACCAUAUAGCUGUUGGGUUGGCCAUCCGGAUUUUGUUAUAAAA CCAUUUCCUCGUGCCUUGACCUUUAACACAUUUGUGAACUUCUUUAAAUCCCUUUU AUUAGUCCUUAAAUACUAAGA
Salivirus FHB	ACAUGGGGGGUCUGCGGACGGCUUCGGCCCACCCGCGACAAGAAUGCCGUCAUC UGUCCUCAUUACCCGUAUUCCUUCCCUUCCCCCGCAACCACCACGCUUACUCGCG CACGUGUUGAGUGGCACGUGCGUUGUCCAAACAGCUACACCCACACCCUUCGGGG CGGGUUUGUCCCGCCCUCGGGUUCCUCGCGGAACCCCCCCCUCCCUCUCUCUCU UUCUAUCCGCCCUCACUUCCCAUAACUACAGUGCUUUGGUAGGUGAGCACCCUGA CCCCCCGCGGAAGCUGCUAACGUGGCAACUGUGGGGAUCCAGGCAGGUUAUCAAA GGCACCCGGUCUUUCCGCCUUCAGGAGUAUCUCUGCCGGUGAAUUCCGGUAGGG CUCUGCUUGGUGCCAACCUCCCCCAAAUGCGCGCUGCGGGAGUGCUCUUCCCCAA CUCAUCUUAGUAACCUCUCAUGUGUGUGCUUGGUCAGCAUAUCUGAGGCGACGUU CCGCUGUCCCAGACCAGUCCAGCAAUGGACGGGCCAGUGUGCGUAGUCGCUUUCC GGUUUUCCGGCGCAUGUUUGGCGAAACGCUGAGGUAAGGUUGGUGUGCCCAACG CCCGUAAUUUGGUGAUACCUCAAGACCACCCAGGAAUGCCAGGGAGGUACCCCAC UUCGGUGGGAUCUGACCCUGGGCUAAUUGUCUACGGUGGUUCUUCUUGCUUCCA CUUCUCUUUUUUCUGGCAUG
EV-D94	UUAAAACAGCCUGUGGGUUGUUCCCACCCCAGAGGCCCACGUGGCGGCCAGUACU CUGGUAUCACGGUACCUUUGUACGCCUGUUUUAUAUCCCCUUCCCCCCGCAACUUA

	GAAGAAAACAAAUCAAGUUCACUAGGAGGGGGUACAAACCAGUACCACCACGAACA AGCACUUCUGUUUCCCCGGUGAUGUCGUAUAGACUGUAACCACGGUUGAAAACGA UUGAUCCGUUAUCCGCUCUUGUACUUCGAAAAGCCCAGUAUCACCUUGGAAUCUU CGAUGCGUUGCGCUCAGCACUCAACCCCAGAGUGUAGCUUAGGUCGAUGAGUCUG GACACUCCUCACCGGCGACGGUGGUCCAGGCUGCGUUGGCGGCCUACCUGUGGU CCAAAGCCACAGGACGCUAGUUGUGAACAAGGUGUGAAGAGCCUAUUGAGCUACA AGAGAAUCCUCCGGCCCCUGAAUGCGGCUAAUCCUAACCACGGAGCAAGGGUACA CAAACCAGUGUAUAUCUUGUCGUAACGCGCAAGUCUGUGGCGGAACCGACUACUU UGGGUGUCCGUGUUUCCUUUUGUUUUUAUCAUGGCUGCUUAUGGUGACAAUCUAA GAUUGUUAUCAUAUAGCUGUUGGAUUGGCCAUCCGGUAAUUUAUUGAGAUUUGAG CAUUUGCUUGUUUCUUCAACAAUUUCACCUAUUCAUUGCAUUUCAGCAGUCAAA
Salivirus NG-J1	UAUGGCAGGCGGGCUUGUGGACGGCUUCGGCCCACCCACAGCAAGAAUGCCAUCA UCUGUCCUCACCCCCAAUUUUCCCUUUUCUUCCCCUGCAACCAUUACGCUUACUC GCAUGUGCAUUGAGUGGUGCAUGUGUUGAACAAACAGCUACACUCACAUGGGGGC GGGUUUUCCCGCCCUACGGCCUCUCGCGAGGCCCACCCCUUCCCUCCCCCUUAUAA CUACAGUGCUUUGGUAGGUAAGCAUCCUGAUCCCCCGCGGAAGCUGCUCACGUGG CAACUGUGGGGACCCAGACAGGUUAUCAAAGGCACCCGGUCUUUCCGCCUUCAGG AGUAUCCCUACUAGUGAAUUCUAGCGGGGCUCUGCUUGGUGCCAACCUCCCCCAA AUGCGCGCUGCGGGAGUGCUCUUCCCCAACUCACCCUAGUAUCCUCUCAUGUGUG UGCUUGGUCAGCAUAUCUGAGACGAUGUUCCGCUGUCCCAGACCAGUCCAGUAAU GGACGGGCCAGUGCGUGUAGUCGUCUUCCGGCUUGUCCGGGGCAUGUUUGGUGA ACCGGUGGGGUAAGGUUGGUGUGCCCAACGCCCGUACUUUGGUGACACCUCAAGA CCACCCAGGAAUGCCAGGGAGGUACCCCACCUCACGGUGGGAUCUGACCCUGGGC UAAUUGUCUACGGUGGUUCUUCUUGCUUCCACUUCUUUCUUCUGUUCACG
Salivirus A strain GUT	UAUGGCAGGCGGGCUUGUGGACGGUUUCGGCCCACCCACAGCAAGAAUGCCAUCA UCUGUCCUCACCCCCAAUUUUCCCUUUCUUCCCCUGCAAUCAUCACGCUUACUCG CAUGUGCAUUGAGUGGUGCAUGUGUUGAACAAACAGCUACACUCACAUGGGGGCG GGUUUUCCCGCCCUACGGCCUCUCGCGAGGCCCACCCUUCCCCUCCCCUUAUAAC UACAGUGCUUUGGCAGGUAAGCAUCCUGAUCCCCCGCGGAAGCUGCUCACGUGGC AACUGUGGGGACCCAGACAGGUUAUCAAAGGCACCCGGUCUUUCCGCCUUCAGGA GCAUCCCCACUAGUGAAUUCUAGUGGGGCUCUGCUUGGUGCCAACCUCCCCCAAA UGCGCGCUGCGGGAGUGCUCUUCCCCAACCCAUCCUAGUAUCCUCUCAUGUGUGU GCUUGGUCAGCAUAUCUGAGACGACGUUCCGCUGUCCCAGACCAGUCCAGUAAUG GACGGGCCAGUGUGCGUAGUCGUCUUCCGGCUUGUCCGGCGCAUGUUUGGUGAA CCGGUGGGGUAAGGUUGGUGUGCCCAACGCCCGUACUUUGGUGACACCUCAAGA CCACCCAGGAAUGCCAGGGAGGUACCCCGCCUCACGGCGGGAUCUGACCCUGGG CUAAUUGUCUACGGUGGUUCUUCUUGCUUCCACUUCUUUCUU
EMCV	UUGCCAGUCUGCUCGAUAUCGCAGGCUGGGUCCGUGACUACCCACUCCCCCUUUC AACGUGAAGGCUACGAUAGUGCCAGGGCGGGUACUGCCGUAAGUGCCACCCCAAA ACAACAACAACCCCCCCUCUCCCUCCCCCCCCCCUAACGUUACUGGCCGAAGCCGC UUGGAAUAAGGCCGGUGUGCGUUUGUCUAUAUGUUAUUUUCCACCAUAUUGCCGU CUUUUGGCAAUGUGAGGGCCCGGAAACCUGGCCCUGUCUUCUUGACGAGCAUUCC UAGGGGUCUUUCCCCUCUCGCCAAAGGAAUGCAAGGUCUGUUGAAUGUCGUGAAG GAAGCAGUUCCUCUGGAAGCUUCUUGAAGACAAACAACGUCUGUAGCGACCCUUU GCAGGCAGCGGAACCCCCCACCUGGCGACAGGUGCCUCUGCGGCCAAAAGCCACG UGUAUAAGAUACACCUGCAAAGGCGGCACAACCCCAGUGCCACGUUGUGAGUUGG AUAGUUGUGGAAAGAGUCAAAUGGCUCUCCUCAAGCGUAUUCAACAAGGGGCUGA AGGAUGCCCAGAAGGUACCCCAUUGUAUGGGAUCUGAUCUGGGGCCUCGGUGCAC AUGCUUUACAUGUGUUUAGUCGAGGUUAAAAAACGUCUAGGCCCCCCUAACCACG GGGACGUGGUUUUCCUUUGAAAAACACGAUGAUAAU
Human echovirus 11	UUGCGGUACCUUAGUGCGCCUGUUUUAUAUACCCGUCCCCCAAACGUAACUUAGA CGCAUGUCAACGAAGACCAAUAGUAAGCGCAGCACACCAGCUGUGUUCCGGUCAA GCACUUCUGUUACCCCGGACCGAGUAUCAAUAAGCUACUCACGUGGCUGAAGGAG AAAACGUUCGUUACCCGACCAAUUACUUCAAGAAACCUAGUAACACCAUGAAGGUU GCGCAGUGUUUCGCUCCGCACAACCCCAGUGUAGAUCAGGUCGAUGAGUCACCGC AUUCCCCACGGGUGACCGUGGCGGUGGCUGCGCUGGCGGCCUGCCCAUGGGGAA ACCCAUGGGACGCUUCAAUACUGACAUGGUGCGAAGAGUCUAUUGAGCUAAUUGG UAGUCCUCCGGCCCCUGAAUGCGGCUAAUCCUAACUGCGGAGCAGAUACCCACAC

	ACCAGUGGGCAGUCUGUCGUAACGGGCAACUCUGCAGCGGAACCGACUACUUUGG GUGUCCGUGUUUCUCUUUAUCCUUAUACUGGCUGCUUAUGGUGACAAUUGAGAGA UUGUUACCAUAUAGCUAUUGGAUUGGCCAUCCGGUGACAAAUAGAGCAAUUGUGU AUUUGUUUGUUGGUUUCGUGCCAUUAAAUUACAAGGUUCUAAACACCCUUAAUCUU AUUAUAGCAUUCAACACAACAAA
SimianV4	UUAAAAUAGCUGCUGGGUUGUUCCCAUCCAGCAGGCCCACUGGGCGUGAUACUCU GGUAUUGCGGUACCUUUGUAUGCCUAUUUUACCUCCCUUCCCCCUGAAACUUAGA AGAAAGAAACAAACACGCUCACUAGGUACACUGCAUCCAGCAGUGUAAUGAGCAAG CACUUCUGUUUCCCCGGAAGGGAUAUAUGGUACGCUGUGCAAACGGCGGAAAUUA AUCCUACCGUUAACCGCCCAUCUACUCCGAGAAGCCUAGUACCUAAUUGAACUUAU CAAUGGAGUUGCGCUCAGCAGGUGACCCUGACCUGCCAGCUCCGGCUGAUGGACC UGGGCAUCCCCCACAGGCGACUGUGGCCCAGGUCGCGUGGCGGCCGGCCCACUC UUUUGAGUGGGACGCCUUGAUAAUGACAAGGUGGGAAGAGCCUAUUGGGCUAGCU GGUUUCCUCCGGCCUCCUGAAUGCGGCUAACCCUAACCCCGGAGCAUAUGGUGGC AACCCAGCCACUGGUAUGUCGUAACGCGUAAGUCUGGGAUGGGACCGACUACUUU GGAGAGUCCGUGUUUCCAUUAUUUUCUUUAUUGAAUCUUAUGGUGACAAAUUGGG AGUUACUCGGGUGACGAUUGAUUAUUACUUCUGAAAACGUAGAGAUAUAGAAUCAA CACA
Salivirus A SZ1	UCUGUCCUCACCCCAUCUUCCCUUCUUUCCUGCACCGUUACGCUUACUCGCAUGU GCAUUGAGUGGUGCACGUGCUUGAACAAACAGCUACACUCACAUGGGGGCGGGUU UUCCCGCCCUGCGGCCUCUCGCGAGGCCCACCCCUCCCCUUCCUCCCAUAACUAC AGUGCUUUGGUAGGUAAGCAUCCUGAUCCCCCGCGGAAGCUGCUCACGUGGCAAC UGUGGGGACCCAGACAGGUUAUCAAAGGCACCCGGUCUUUCCGCCUUCAGGAGUA UCCCUGCUAGUGAAUUCUAGUAGGGCUCUGCUUGGUGCCAACCUCCCCCAAAUGC GCGCUGCGGGAGUGCUCUUCCCCAACUCACCCUAGUAUCCUCUCAUGUGUGUGCU UGGUCAGCAUAUCUGAGACGAUGUUCCGCUGUCCCAGACCAGUCCAGUAAUGGAC GGGCCAGUGUGCGUAGUCGUCUUCCGGCUUGUCCGGCGCAUGUUUGGUGAACCG GUGGGGUAAGGUUGGUGUGCCCAACGCCCGUACUUUGGUGAUACCUCAAGACCAC CCAGGAAUGCCAGGGAGGUACCCCGCUUCACAGCGGGAUCUGACCCUGGGCUAAU UGUCUACGGUGGUUCUUCUUGCUUCCACUUCUUUCUACUGUUCAUG
Phopivirus	GGGAGUAAACCUCACCACCGUUUGCCGUGGUUUACGGCUACCUAUUUUUGGAUGU AAAUAUUAAUUCCUGCAGGUUCAGGUCUCUUGAAUUAUGUCCACGCUAGUGGCAC UCUCUUACCCAUAAGUGACGCCUUAGCGGAACCUUUCUACACUUGAUGUGGUUAG GGGUUACAUUAUUUCCCUGGGCCUUCUUUGGCCCUUUUUCCCCUGCACUAUCAUU CUUUCUUCCGGGCUCUCAGCAUGCCAAUGUUCCGACCGGUGCGCCCGCCGGGGU UAACUCCAUGGUUAGCAUGGAGCUGUAGGCCCUAAAAGUGCUGACACUGGAACUG GACUAUUGAAGCAUACACUGUUAACUGAAACAUGUAACUCCAAUCGAUCUUCUACA AGGGGUAGGCUACGGGUGAAACCCCUUAGGUUAAUACUCAUAUUGAGAGAUACUU CUGAUAGGUUAAGGUUGCUGGAUAAUGGUGAGUUUAACGACAAAAACCAUUCAACA GCUGUGGGCCAACCUCAUCAGGUAGAUGCUUUUGGAGCCAAGUGCGUAGGGGUG UGUGUGGAAAUGCUUCAGUGGAAGGUGCCCUCCCGAAAGGUCGUAGGGGUAAUCA GGGGCAGUUAGGUUUCCACAAUUACAAUUUGAA
HRV-B37	UUUAAACAGCGGAUGGGUAUCCCACCAUCCGACCCACUGGGUGUAGUACUCUGGU AUUUUGUACCUUUGUACGCCUGUUGUUCCUAAUGUACCCACCCUAAAACUUCCUAC CCAAGUAACGUUAGAAGUUUCAUCAACAAGUACAAUAGGAAGCAUCACAUCCAGUG GUGUUUUGUACAAGCACUUCUGUUUCCCCGGAGCGAGGUAUAGGCUGUACCCACU GCCGAAAGCCUUUAACCGUUAUCCGCCAACCAACUACGUAAAAGCUAGUAUCAUCA UGUUUUAAAAUAGGCGUUCGAUCAGGUGGUACCCCCCUCCACUAGUUUGGUCGAU GAGGCUAGGAACUCCCCACGGGUGACCGUGUCCUAGCCUGCGUGGCGGCCAACC CAGCUUCUGCUGGGACGCCUUUUUAUGGACAUGGUGUGAAGACUCGCAUGUGCUU GGUUGUGACUCCUCCGGCCCCUGAAUGCGGCUAACCUUAACCCCGGAGCCCUGUG UUGCAAUCCAGUAACAUUAGGGUCGUAAUGAGCAAUUUCGGGACGGGACCGACUA CUUUGGGUGUCCGUGUUUCUCAUUUUUCUUAUUAUUGUCUUAUGGUCACAGCAUA UAUAUAACGUAUAUACUGUGAUC
HRVB93	UUAAAACAGUGGAUGGGUUUCCUACCAUCCGACCCACUGGGUGUAGUGCUCUGGU AUAUUGUACCUUUGCACGCCUGUUUUCCCAUUGUACCCUUCCUUAAAUUCCUCCC CAUGUAACGUUAGAAGUUUAAGAACAUAAAUGUACAAUAGGAAGCAUCACAUCCAG UGGUGUAUGGUACAAGCACUUCUGUUUAUCCGGAGCGAGGUAUAAGCGGUACCCA

	CUGCUGAAAGCCUUUAACCGUUAACCGCCAAUCAACUACGUAACGGCUAGUAUUAC CUUGUUUGCAAUCUGGCGUUCGAUCAGGUGGUUCCCCCCACUAGUUUGGUCGAU GAGGCUAGGAGUUCCCCACGGGCGACCGUGUCCUAGCCUGCGUGGCGGCCAACC CAGCAUUUGCUGGGACGCCUUUUUGUAGACAUGGUGUGAAGACCUGCAUGUGCUU GAUUGUGAGUCCUCCGGCCCCUGAAUGCGGCUAACCUUAACCCCGGAGCCUUGUG ACAUAAUCCAAUGUCAUUAAGGUCGUAAUGAGCAAUUCUGGGACGGGACCGACUA CUUUGGGUGUCCGUGUUUCUCUUUAUUCUUUACAUUGUCUUAUGGUCACAGCAUA UAUAAUACAUAUAUACUGUGAUC
HRV-A1	UUAAAACUGGGUGUGGGUUGUUCCCACCCACACCACCCAAUGGGUGUUGUACUCU GUUAUUCCGGUAACUUUGUACGCCAGUUUUUCCCUCCCCUCCCCAUCCUUUUACG UAACUUAGAAGUUUUAAAUACAAGACCAAUAGUAGGCAACUCUCCAGGUUGUCUAA GGUCAAGCACUUCUGUUUCCCCGGUUGAUGUUGAUAUGCUCCAACAGGGCAAAAA CAACAGAUACCGUUAUCCGCAAAGUGCCUACACAGAGCUUAGUAGGAUUCUGAAAG AUCUUUGGUUGGUCGUUCAGCUGCAUACCCAGCAGUAGACCUUGCAGAUGAGGCU GGACAUUCCCCACUGGUAACAGUGGUCCAGCCUGCGUGGCUGCCUGCGCACCUCU CAUGAGGUGUGAAGCCAAAGAUCGGACAGGGUGUGAAGAGCCGCGUGUGCUCACU UUGAGUCCUCCGGCCCCUGAAUGCGGCUAACCUUAAACCUGCAGCCAUGGCUCAU AAGCCAAUGAGUUUAUGGUCGUAACGAGUAAUUGCGGGAUGGGACCGACUACUUU GGGUGUCCGUGUUUCACUUUUUCCUUUAUUAAUUGCUUAUGGUGACAAUAUAUAU AUUGAUAUAUAUUGGCAUC
HRVB3	UUAAAACAGCGGAUGGGUACCCCACCAUCCGACCCACUGGGUGUAGUACUCUGGU ACUUCGUACCUUUGUACGCCUGUUCUUCCCAUUGUACCCUUCCUGAACUUCCAAC CCAAGUAACGUUAGAAGCUCAACAUUUAGUACAACAGGAAGCACCACAUCCAGUGG UGUUUAGUACAAGCACUUCUGUUUCCCCGGAGCGAGGUAUAGGCUGUACCCACUG CCAAAAACCUUUAACCGUUAUCCGCCAACCAACUACGUAAAAGCUAGUAGUAUUAU GUUUUUAACUAGGCGUUCGAUCAGGUGGAUUUCCCCUCCACUAGUUUGGUCGAUG AGGCUAGGAAUUCCCCACGGGUGACCGUGUCCUAGCCUGCGUGGCGGCCAACCCA GCUUAUGCUGGGACGCCUUUUUAUAGACAUGGUGUGAAGACUCGCAUGUGCUUGG UUGUGAUUCCUCCGGCCCCUGAAUGCGGCUAACCUUAACCCUGGAGCCUUGUGUC ACAAACCAGUGAUGAUAAGGUCGUAAUGAGCAAUUCCGGGACGGGACCGACUACU UUGGGUGUCCGUGUUUCUUAUUUUUCUUAUUAUUGUCUUAUGGUCACAGCAUAUA UAUAACAUAUACUGUGAUC
HRVB4	UUAAAACAGCGGAUGGGUUUCCCACCAUCCGACCCACUGGGUGUAGUGCUCUGGU AUUUUGUACCUUUGCACGCCUGUUUCCCAUUUGUACCCUUCCUUAAUCUCCUUCC CCCGUAACGUUAGAAGUUUUGGAAUUUUAAAGUACAAUAGGAAGCGCCACAUCCAG UGGUGUUGCGUACAAAUACUUCUGUUUACCCGGAGCGAGGUAUAGGUUGUACCCA CGGCCAAAAGCCUUUAACCGUUAUCCGCCAAUCAACUACGUAACGGCUAGUAUCAU CUUGCUUUUGAUUUGGUGUUCGAUCAGGUGGUAUCCCCCACUAGUCUGGUCGAU GAGGCUAGGAAUUCCCCACGGGCGACCGUGUCCUAGCCUGCGUGGCGGCCAGCC CAGCUUUUGCUGGGACGCCUUUUCAAAGACAUGGUGUGAAGACCUGCAUGUGCUU GGUUGUGAGUCCUCCGGCCCCUGAAUGCGGCUAACCUUAACCCUGGAGCCCAGCA GCAUAAUCCAAUGUUGUUUGGGUCGUAAUGAGCAAUUCCGGGACGGGACCGACUA CUUUGGGUGUCCGUGUUUCCUUUUAUUCUUACAUUGUCUUAUGGUCACAGCAUAU AUAUUAUAUAUACUGUGAUC
HRV-A89	UUAAAACUGGGAGUGGGUUGUUCCCACUCACUCCACCCAUGCGGUGUUGUACUCU GUUAUUACGGUAACUUUGUACGCCAGUUUUUCCCACCCUUCCCCCAUAAUGUAACU UAGAAGUUUGUACAAUAUGACCAAUAGGUGACAAUCAUCCAGACUGUCAAAGGUCA AGCACUUCUGUUUCCCCGGUCAAUGAGGAUAUGCUUUACCCAAGGCAAAAACCUUA GAGAUCGUUAUCCCCACACUGCCUACACAGAGCCCAGUACCAUUUUUGAUAUAAUU GGGUUGGUCGCUCCCUGCAAACCCAGCAGUAGACCUGGCAGAUGAGGCUGGACAU UCCCCACUGGCGACAGUGGUCCAGCCUGCGUGGCUGCCUGCUCACCCUUCUUGG GUGAGAAGCCUAAUUAUUGACAAGGUGUGAAGAGCCGCGUGUGCUCAGUGUGCUU CCUCCGGCCCCUGAAUGUGGCUAACCUUAACCCUGCAGCCGUUGCCCAUAAUCCA AUGGGUUUGCGGUCGUAAUGCGUAAGUGCGGGAUGGGACCAACUACUUUGGGUG UCCGUGUUUCCUGUUUUUCUUUUGAUUGCAUUUUAUGGUGACAAUUUAUAGUGUA UAGAUUGUCAUC
HRV-C54	UUAAAACAGCUGUGUGGUUGUUCCCACCACCAGGCACACUGUGCGUUGUACACUG GGAUUCCGGUCACUUUGUACGCCUGUUUGCUAUCCCCCCCAACUUAUGUAAUUUA

	GAAGAUGUACACAACGCCCAUUAGGAUGCGGCCAAACCAGGUCCGCUUAGGGCAA GCACUUCUGUUUCCCCGGGUGUGUGAAUAGACUCUAACAGGGUUGAAGCUGUAGC ACUCGUUAUCCGCGCAACUACUACGCGAAUGUUAGUAGCAUCCUGUGUUGCAUUU GGGAUUUCGCUCCGCAGAAAACCCCAUCUGUAGAUUAGGGCAAUGAGGCUACACA UACCCCACUGGCGACAGUGGUGUAGCCUGCGUGGUGCCCUACCCAGGCCAUCUU GGCCUGGGAUUCCACUUACAAGACAGGGUGUGAAGGCACUAGUGUGCUAGUUGUG AGUCCUCCGGCCCCUGAAUGCGGCUAAUCUUAACCCCGUAGCCCCCGCAAGUAAA CCAACUUGUAGGUGGUCGUAAUGAGUAAUUACGGGAUGGAACCGACUACUUUGGG UGUCCGUGUUUCCUUUUAUUCUUUAUAUUUGCAUCCUAUGGUUACAACAUAAGUAA UC
Crohivirus B	GUAUAAGAGACAGGUGUUUGCCUUGUCUUCGGACUGGCAUCUUGGGACCAACCCC CCUUUUCCCCAGCCAUGGGUUAAAUGGCAAUAAAGGACGUAACAACUUUGUAACCA UUAAGCUUUGUAAUUUUGUAACCACUAAGCUUUGUGCACAUAAUGUAACCAUCAAG CUUGUUAGUCCCAGCAGGAGGUUUGCAUGCUUGUAGCCGAAAUGGGGCUCGACCC CCCAUAGUAGGAUACUUGAUUUUGCAUUCCAUUGUGGACCUGCAAACUCUACACAU AGAGGCUUUGUCUUGCAUCUAAACACCUGAGUACAGUGUGUACCUAGACCCUAUA GUACGGGAGGACCGUUUGUUUCCUCAAUAACCCUACAUAAUAGGCUAGGUGGGCA UGCCCAAUUUGCAAGAUCCCAGACUGGGGGUCGGUCUGGGCAGGGUUAGAUCCC UGUUAGCUACUGCCUGAUAGGGUGGUGCUCAACCAUGUGUAGUUUAAAUUGAGCU GUUCAUAUACC
Sapelovirus	CCCCCCUCCACCCUUAAGGUGGUUGUAUCCCCUACACCCUACCCUCCCUUCCACA UAGGACGAAUAAACGGACUUGAGAUUAAGGCAAGUACAUAAGGUAUGGUUUUUGG AUACACUUAAAUGGCAGUAGCGUGGCGAGCUAUGGAAAAAUCGCAAUUGUCGAUA GCCAUGUUAGCGACGCGCUUCGGCGUGCUCCUUUGGUGAUUCGGCGACUGGUUA CAGGAGAGUAGACAGUGAGCUAUGGGCAAACCCCUACAGUAUUACUUAGGGGAAU GUGCAAUUGAGACUUGACGAGCGUCUCUUUGAGAUGUGGCGCAUGCUCUUGGCAU UACCAUAGUGAGCUUCCAGGUUGGGAAACCUGGACUGGGUCUAUACUGCCUGAUA GGGUCGCGGCUGGCCGCCUGUAACUAGUAUAGUCAGUUGAAAACCCCCC
HCV	GCCAGCCCCCGAUUGGGGGCGACACUCCACCAUAGAUCACUCCCCUGUGAGGAAC UACUGUCUUCACGCAGAAAGCGUCUAGCCAUGGCGUUAGUAUGAGAGUCGUGCAG CCUCCAGGACCCCCCCUCCCGGGAGAGCCAUAGUGGUCUGCGGAACCGGUGAGUA CACCGGAAUUGCCAGGACGACCGGGUCCUUUCUUGGAUCAACCCGCUCAAUGCCU GGAGAUUUGGGCGUGCCCCCGCAAGACUGCUAGCCGAGUAGUGUUGGGUCGCGA AAGGCCUUGUGGUACUGCCUGAUAGGGUGCUUGCGAGUGCCCCGGGAGGUCUCG UAGACCGUGCACCAUGAGCACGAAUCCUAAACCUCAAAGAAAAACCAAACGUAAC
Covid19	AUUAAAGGUUUAUACCUUCCCAGGUAACAAACCAACCAACUUUCGAUCUCUUGUAG AUCUGUUCUCUAAACGAACUUUAAAAUCUGUGUGGCUGUCACUCGGCUGCAUGCU UAGUGCACUCACGCAGUAUAAUUAAUAACUAAUUACUGUCGUUGACAGGACACGAG UAACUCGUCUAUCUUCUGCAGGCUGCUUACGGUUUCGUCCGUGUUGCAGCCGAUC AUCAGCACAUCUAGGUUUCGUCCGGGUGUGACCGAAAGGUAAG
CrPV	UUCUAGCACUAGUAAAGCAAAAAUGUGAUCUUGCUUGUAAAUACAAUUUUGAGAGG UUAAUAAAUUACAAGUAGUGCUAUUUUUGUAUUUAGGUUAGCUAUUUAGCUUUACG UUCCAGGAUGCCUAGUGGCAGCCCCACAAUAUCCAGGAAGCCCUCUCUGCGGUUU UUCAGAUUAGGUAGUCGAAAAACCUAAGAAAUUUACCUGCUACAUUUCAAGAUAAU CUAGCCACC

Supplementary Table 2 | CITE and H2SL sequences.

Name	Sequence (5'-3')
PEMV2	GGCUUCGCUUCCCGCCGGAAGACCGCGGCGGUUCUGUUCCUCCCACAGGAGUAC GGCAACAACCCACCUUGGGAAAGUGGGGACCCCAGCACUAACUCCUUUAACUAGG CGGGCGUGCUGGUUACAGUAGGAGGGGACAGUGCGCAUCAAAACUGAGCCCCACC ACAACUCUCAUCCACAGGGUGGUUGGGACGCAGGUGUCGGAGGGAUCGCCAGCCC UCAGGAUAGUGAGCUCCCGCAGAGGGAUAAGCUAUCUCCCUGCGACGUAGUGGUA GAACACGUGGGAUAGGGGAUGACCUUGUCGACCGGUUAUCGGUCCCCUGCUCCUU CGAGCUGGCAAGGCGCCCAAAGGUUCUACACUGCUACUAAAGUUGGUGGUGGAUG UCUCGCCCAAAAAGAUCAUAAACGCGCGGGAUAAGGUCCCUUUCCACCUUCGCCG GGUAAGGCUAGAGUCAGCGCUGCAUGACUAUAACUUGCGGCCGAUCCAGUUGCAC GACUGGUGGUCCCCCUCAGUGUCUUGGUUGUCUGCCAAGUGGGCGGUGGUCGGA UUCCACCACACCCUGCCACGAGGUGCGUGGAGACUUGGCCAGUCUAGGCUCGUCG UAAUUAGUUGCAGCGACGUUAAUCAACCCGUCCGGGCAUACAAUAGGACCGGUUG UGCUUCUUCCUCCCUUUCUUAGCCAGGGUGGUUACCUCCCUGGCGCCC
RCNMV1	UUGGUUCUUUUAAGUGUAGCCUCCACCCGAGUUGCAAGAGGGGAACACGCAGUCU CGCCGACCCUGUUGGCAAACAGUAAAAUUGCAAAAAAUAGAGUGCUAGGAGUAGUU CCCGUACCCGCGGGAGCAAGACCCUACUACAGUAGACGAACCGGCAUCGGACCCU GGGAAACAGGUACCUAGCGUAUUAAUAGGUCGCUAGAAAUCGGGCGGGCUUCACC UUCGGGUAACCCGCCGUCGAGAGGGAAAACCUCUGUGUGGUUGUGCGCACGUCG UUGUCGGAAGACUCUCACUGUUGUAGUUAUUUCCUUUUUCUUUUAGGUAGGAGCA CUUGCUCAGAGGGCGCAUACCUCUGGUAAAACAAAAUUGGCCCCUGAGAGUAGAAA ACUCAGGUUAAUAAAACAGGAGAAUAUUUACCUAGUCGUAUAACGGCUAGGUACCC C
CIRV	GUUCGCUGAAACGAGUGUAAAUCUGGCAUAGCAUACAGGUUACUCUUGUUGGGUU CUAGAUGUUAUGAUGACGAGUCGGUUCGGGCUCCGCACUAGGUUUGGUCACCUAG GGGAUGGAGAUAUGGAAAGGGUCUCGUGUGCUGUUAGACGGUCGUAAGACGCGC UUGCAACAUGGGCCUAUGACCGGAUAAGUCAUAGCAAUACUAGCCAACAUGAAUAG GAUUCCUGUUUACGAAAGUUAGAUGUCACUUGUGGAAACGGACCCAGACACGGUU GAUCUCACCUUCGGGGGGGCUAUAGAGAUCGCUGGAAGCACUACCGGACAACCGG AACAUUGCAGAAAUGCAGCCC
TBSV	GUUUGUGGAGAUGAGUGUAAAUCUGGCAUAGCAUACAGGUUACUCUUGUUGGGUU CUGGAUGUUAGGAUGACGAGUCGACUCGGGCUCCGCACUAGGUUUGGUCGCCUA GGGGAUGGAGAUAUGGAAAGGGUCUCGUGUGGUAUCAGUCGGUCGAAAGACGCG CUUCCAACAUGGGCCUAUGGUCGGAUAAGUCUUAGCAAUACCAGCCAGCAUGAAUU GGAUUCCUGUUUACGAAAGUUAGGUGUCACUUGUGGAAGCGGACCCAGACACGGU UGAUCUCACCCUUCGGGGGGCUAUAGAGAUCGCUGGAAGCACUACCGGACAACCG GAACAUUGCAGAAAUGCAGCCC
PMV	ACCUGCCCAAGGUUAUCACAGGCUAGGACCGGCCUGAUAAAGGUCCGACUUGAUG UGCAGCUUCUUAGGUGCAUCUCGUAUAGAAGCGCUCUGGACAUGACACCUUUGAG ACAGACCGUACAGCAGUCACACGGGACGCCACACCACCUUUGCAGAGGUGCCCUU GGGAAACCAAUGGUGUGGGGUGACACUGAUUAGUCGUUAACGGUGUACUAAUCCA GAGUAGUUCAGUGCAGGGGAAACCCGGGCCGUAAGCACGUGGAACUAUAUAACUU AAUCUCGCUGAGAAGAUUGGGGGGGAUUCAUGAAAUCCCCUCACCAGGCCC
TPAV	ACUCGCUCAGAGUCAUCCUAGGCUAGGACCGGCCAGACAAAGGUCCGACACUACA UGCAGCGCUCUAGGUGUGUAGUGACAGUAAGCGCGCUCUAUCCGAAACUCCAGUG ACAGGACAAGACCCCAGGCGGGACGCCACACCCAUGUUUGCAGACAUGGCCCGAG GAAACUCUGGUGUGGGCUGGGGGGAAUCCUAUGAUCACUGGACGGAAGGUAGAGA GUUCCAGCACAACCCCUUUAGGGUGGGCCGUAAGUGCGUGGGAACAUUUCCUACA GUCUCACUGAGAAGAUUGGACGGGGUACCAUGAGUACCCCUCACCAGGCCC
MNSV-264	AUCUCGCCUUAGGUUUACGGUUACCCGAGCUUCGCCUGAAACGGCGUUCCUCUAA UUCCUUCCUGGUUCUUUCCCUCCACGGCUCGCGUGUUCUGCGGAACUGCAUCUGC UUCACUUCCUUCUUUUUCCCUCUGUGGUGGUACGACCAAUGGGUGGCUCCCAUUG ACUGCGUAGGACAUGGAAAAAGAUUGAGUGGAGUGCACGAAGCGUCGCAAGAUCG CUGCCUGACGAGGUCCAGCCAAUCAAAUACCAAACAUCUGGUUUGGGGGCGGAUU GGCUGUAUGUCAUGUAAGUUUGAGGGGUUCUAUCGGAGCCCCCCGCCC

TNV-D	UUGCUUUCAUAGACCCGUCUUCCGGGAGACGUUAAGAAGAUGCCCGUGGAAAAUA GUAGGUAGAGUUGUGGCGUGACAACCCACAACUGCAUCUCUUUGCUUAUCUAGUU ACAAUAUAUGUUGACGUACAAGAAGGAUCCUGGGAAACAGGUUCAACGGGCUCACU GUGGUGGUGAGCCGUCGCAUCACUUGUAUGUGCUCCAAUAUUGGUUGUCGAUAAG CGUCCUGACAUGGCUCCAUGCAACAGCAUGGGGGGGUCCAGAGUCAGUCCCCUCU UUAUUUACCUAGAGAUUCCUCUAGGAAUU
CarMV	AUGCAAGAGAAAAGUUUGGUAGACACACACGGCGGGACGUGUUGCUAGAGCUGCC ACUCUGCCCGAGAUGGAUCGAUAUUAACCAUCUUUGGCAACAGCGUGUGUGAUCG ACCAAACCUAGUAAUUGCGUAGCAUCCGACUGUAACCUGUUCUCUGCUGAAUGCGA GGGUGGCCUCGUGAUCCCAAACCAUAAAAGAAACCAACUUGUUGGUGGAAGGGCU UGGAAGUAGGAUCAACCACUGUUUAUUAAUUAGGGGUGAGAGUAUUCACUGUACU CUUUCCCCGCCC
HCRSV	GUUUGUAUUGUAUGGGCCUCCGUGAAGUUAGGGGCAGUGGAAACGUCGGUCUAG CCAGCCGUCCCCUGGGUAGUGUGCUCUGUCUAUGUACACCACUACUCGGUUUCCA CAAACGAUCGGACUCCCAAUAUAGUGUACAUUAUGUUGAUUGCUUCUUUCUACCUA UAUAAUAUCUCCUCUCAUCUCUCCUCUACGCUUGGAAUCCAUAAAAUCCUUCGGGU GGGCUUGGAUUUCGGGUACUUGUUUGAAAUUCCCUCACGACCGCCAUAGUUGUGA GGCAGCCC
HIF-1α	GUGCUGCCUCGUCUGAGGGGACAGGAGGAUCACCCUCUUCGUCGCUUCGGCCAG UGUGUCGGGCUGGGCCCUGACAAGCCACCUGAGGAGAGGCUCGGAGCCGGGCCC GGACCCCGGCGAUUGCCGCCCGCUUCUCUCUAGUCUCACGAGGGGUUUCCCGCC UCGCACCCCCACCUCUGGACUUGCCUUUCCUUCUCUUCUCCGCGUGUGGAGGGAG CCAGCGCUUAGGCCGGAGCGAGCCUGGGGGCCGCCCGCCGUGAAGACAUCGCGG GGACCGAUUCACC
PCRPV	UUAUUUGUAUAUGUUAUGUAUAAUUAUAAUCAUGCAAUGGCACAUGUAGUUCCCCU AACCUCUCAGUAGGAGUCAAGCCCUGAUGCGCCAACAUCCAGAAGGGGGUAAAGC UGAGUCGGUGCAAGGAACCUUGUUAGCCCUAAGAAAAACCCAAAAACAGAGUGCAU AGUUUAGCUUUUGUGUUGUAGGAGCCAGAGACCAAAGAUCCAGACAACUGGUGAG GGCGGUCGAUGGUGCGAAGUAAAUUAGAAAGGGGGGCGUUUACCCACGUCCUCCC UUGCCC
CCFV	AGUAAGUAGCCUAACCUGCUGCAGCAGGAAGUAUUCUAGGUAGUAUAGUAACUCUU UCUUCAACUUCUUACUUAGAAAAACACAAAAAUACCCCCAGCUCGCUUUUGUUUUG AGCUGUGUUAGGGACCAAAAACGGCGACUCAUCCGUCCUGAGUGGCCUCUGGUGU GGAAACCAGAGAGUCUUGCGGUAACCAACGAAAUCUCUUCGGAGUGGGUGUUGGG GACCACCAAAUGAAGAAAAUAGCCCCUCCCUCGCGCAGGGGGGGGCCAGCCC
PSNV	CAAUAUGCACGAGCUGUCUUGUAACACAGCAAUGGAGUAAGUGUACUCUGGUAGC AAUGGGUGGGGUCGAUUUCCUACGUCUAGCUUUGGCCGGCUAGCGUCUCUCGUA GUGGGGAAAUAACUCAUUGAGACCAGCUCAGAACGCUGACAUUGUCUGUGUUUGG AUAGAGGGCUUGGGGAAAACUCCGUAAGGAGGGGCGCCCGAGAGCCUAGUAACAC CAAAUUUAUAAACUAGGGGGGACCGUUAAGGUCUCCCCCGCCCC
TCV	UACGGUAAUAGUGUAGUCUUCUCAUCUUAGUAGUUAGCUCUCUCUUAUAUUAAGAA AAGAAAACAAAAACCCCCAGGUCGCUUUAUUUUGACCUGUGUUAGGGACCAAAAAC GGUGGCAGCACUGUCUAGCUGCGGGCAUUAGCCUGGAAAACUAGUGCUCUCUGGG UAACCACUAAAAUCCCGAAAGGGUGGGCUUUGGGGACCCUCCGAACCAAUAGAUAG CCUCCCUCCUCGGACGGGGGGCCUGCCC
HnRSV	ACGGGAUGAGCUUCUGGGGGCUGCCACCCUCGCCUCAUGCGCAUUGCCUGCUCU GGGCGCAUGUGGAAGCUGAUAUAGUUGAGAGCUUCGUGGAAGGAUGUGGAAACCA AGGGAGGGAGAUAUCCAGGGAGGAUGUCGGAGUUGUAAGGAUAAAAUACCUAAGC UUCAUCAGGUUAGGUUGGGCUUCCGAUGCAACAAAACACCAGUAGAUAAAUAGGG GGGGAUCCAGGGGAUCCUUCUCCGCCC
PLPV	GGAAAGCCACUGCGGUGGGGUCAGUCUCCUGGUGCGCCAACACCCAGAAGGGGAG ACAACUUCAGUGGUUCAAAACCCGAUGCUUCGUGUAAGCCUAGAAAAAUCCAAAAA CAGAGUGCUACUGCUUUCAGUGUGCGAGCUUUAUUUGCUCACUUCCUCCGUCGAU AAAAUCCCGAAUUUUCGGGUGGGCUUCGGCGGACAUCCUUAAUGAAGAACGGGGG GGUAACCCACCCCCCUGAUCUGCCC

SCV (PTE-TSS)	GUUGCUCGACUGUGAGGGGACCUACCCACUGUGCUGCCACACAGGAACUUCCAAC CUUCGGGUGGCGAGGUAGGGCAGAAGAGUGACUGAACGAGGAUGGUGUAGAGCAA UGGAUGGGGGGGGGAGUCCAUGGUGGGCUCCGGCGCGACGAUGCUAAAAUCUCCA UAUCAUGUGUGGAGUGGGCUGCUUAGUCGCUUAUUAGUUGAAAAUUAGGGGGGUC CUUAGGGACCUCUCCCGCCC
PFBV	GAGGGCAGUCACCCUACCUUCUGCCAGAGGUUCCCGGAGGAGUAGUGAUCCUUUG GGGGUGAUUCAUAGCUAGGAAAGGUCGGAGAGUUGGGAGGAGGCCAACUGGGGG AGGUGCUAGGAGCACCAGAUGCAGCAGAGCAAAAAUCCGGGUUUCAUCGCCUGGU GCGGGCGCUCAAUGCUGCUAACACGUUCCUGAAGGAAGAAAGGUUUCGGUCGAUG GAGACCUUAACCCGCCC
GaMV	AGGAAGAUGCUUGGGGGUGUUCCCCUAAAGUUCAGAAUGGUCUAUCAGGGAGCAG GCACUUUUCCGGCUUUGGCGAGUCGUCUCCACUUGUAUACGAACAGUGAGGGGAA AAGUUAUCUGAUAGAAGAAUGAACCAUGGACAAGGACACUCUAAACCACCACUAAG AAGCUAUAGGGAAACCUUGCUGAGGGCAGUGAGUGGUCGUACCAGAACGAAUAUG GCUUUGCCCCGCCC
CbMV	ACACUAAGAGUUACUGCCACCAAAAGAGGAUAAUCAGACCACUUGUAGUUUGCUGC CACAAACUGGUUGACAAGUUAAUGCCUCUCUCUUAGUUAGUAAGCCGGAUUCAAGU AGGGAAUGUGGAGGGAGAUGCCAUGGUGGCAUCAGAGCAGACCUGGAAAUACUCU UCGGAGGGGCGCCAGAACUGCACUAAACACUUGUUAUCAGGGGACUGUUGAGGAG UCUCCCCGCCCG
MNeSV	GACAAACUCUCCGCCUAAUGUCUGAUGAUGUGAGGAACGUGGACUGUGAUGUGGU GGUGCGGUACCAUGGCUGGUCACCAUGGUAAUGCGUAGGGCAACACAGUUCAUUA AGACUCACUGAUGAUGGCACUAGGCACGGUUCACCCCCAUCCUUCGGGAGGGCUA UAGGGGGUGACCGGGUUACACCACCGGAAGACCGGAACAUUGCCUUUGGGCAGCCC C
FGF9	UGAAACAGCAGAUUACUUUUAUUUAUGCAUUUAAUGGAUUGAAGAAAAGAACCUUU UUUUUCUCUCUCUCUCUGCAACUGCAGUAAGGGAGGGGAGUUGGAUAUACCUCGC CUAAUAUCUCCUGGGUUGACACCAUCAUUAUUGUUUAUUCUUGUGCUCCAAAAGCC GAGUCCUCUG
p53A	UCUAGAGCCACCGUCCAGGGAGCAGGUAGCUGCUGGGCUCCGGGGACACUUUGC GUUCGGGCUGGGAGCGUGCUUUCCACGACGGUGACACGCUUCCCUGGAUUGGCA GCCAGACUGCCUUCCGGGUCACUGCC
STNV	ACCAUGGUCAUGAUGAUGAUAGUGAAGGACGCUGAAAGAUGCGUAGCUACCCUCC UGGUGCACUUCCUGGUGCAAAGCAGAACCAAAGGGUACGGUGGUACGGCGGACAG UAGUCCUGAA
p53B	AUGGAGGAGCCGCAGUCAGAUCCUAGCGUCGAGCCCCCUCUGAGUCAGGAAACAU UUUCAGACCUAUGGAAACUACUUCCUGAAAACAACGUUCUGUCCCCCUUGCCGUCC CAAGCA
BYDV	AGACAACACCACUAGCACAAAUCGGAUCCUGGGAAACAGGCAGAACUUCGGUUCGU AAGCUCGGGUAGGCUGUCAACCUACCGCCGUAUCGUAUUGUGUUUGGC
PEMV2 (PTE)	GGUAGAACACGUGGGAUAGGGGAUGACCUUGUCGACCGGUUAUCGGUCCCCUGCU CCUUCGAGCUGGCAAGGCGCCCAAAGGUUCUAC
SCV (PTE)	GUUGCUCGACUGUGAGGGGACCUACCCACUGUGCUGCCACACAGGAACUUCCAAC CUUCGGGUGGCGAGGUAGGGCAGAAGAGUGAC
BBV	GUUUUCGAAACAAAUAAAACAGAAAAGCGAACCUAAACA
H2SL	caaaaggcucuuuucagagccaccca

Supplementary Table 3 | Typical RNA and DNA sequences information.

Feature	Schematic diagram or/and sequence (5'-3')
External-Homology-arm-intron-Internal-homology-arm-BBV-OVA$_{peptide}$-P2A-Internal-homology-arm-intron-External-homology-arm	 GAAUUCCGUACUGCUAGCUCGCUGCCAUCGAUGCUGAUGAGGCACUGGCUGACA GACCGUCGAUUGUCCACUGGUCAAGUAUAUUGAUUAGUUUUGGAGUACUCGAUC AGACGACAAACGUUUUCGAAACAAAUAAAACAGAAAAGCGAACCUAAACACGCCAC CAUGCUUGAGCAGCUUGAGAGUAUAAUCAACUUUGAAAAACUGACUGAAUGGACC AGUGCAGAGAGCCUGAAGAUAUCUCAAGCUGUCCAUGCAGCACAUGCAGAAAUCA AUGAAGCAGGCAGAGAGGUGGUAGGGUCAGCUACUAACUUCAGCCUGCUGAAGC AGGCUGGAGACGUGGAGGAGAACCCUGGACCUUUGUCGUCUGAUCUCUCUAAAU AGCAAUAUUUACCUUUGGAGGGAAAAGUUAUCAGGCAUGCACCUGGUAGCUAGU CUUUAAACCAAUAGAUUGCAUCGGUUUAAAAGGCAAGACCGUCAAAUUGCGGGAA AGGGGUCAACAGCCGUUCAGUACCAAGUCUCAGGGGAAACUUUGAGAUGGCCUU GCAAAGGGUAUGGUAAUAAGCUGACGGACAUGGUCCUAACCACGCAGCCAAGUC CUAAGUCAACAGAUCUUCUGUUGAUAUGGAUGCAGUUCACAGACUAAAUGUCGGU CGGGGAAGAUGUAUUCUUCUCAUAAGAUAUAGUCGGACCUCUCCUUAAUGGGAG CUAGCGGAUGAAGUGAUGCAACACUGGAGCCGCUGGGAACUAAUUUGUAUGCGA GACCAGUGGACAAUCGACGGUCUGUCAGCCAGUGCCUCAUCAGCAUCGAUGGCA GCGAGCUAGCAGUACG
RzL-ribozyme-BBV-3×Flag-P2A-Substrate	 GAGACCGCAACUGAAAAGUUGUUAUCACUUGUCGUAAGACACUUUGGAUGGGUU GAAGAACUGCACGUAGAAGCAAAAGGCACGUUUUCGAAACAAAUAAAACAGAAAAG CGAACCUAAACACGCCACCAUGGACUACAAGGAUCAUGACGGAGACUAUAAGGAU CACGAUAUUGAUUACAAAGAUGACGACGACAAAGCUACUAACUUCAGCCUGCUGA AGCAGGCUGGAGACGUGGAGGAGAACCCUGGACCUGUGAGACCAAGAAACGUGC AGUUU
MOG_{35-55}	AUGGAAGUGGGCUGGUAUCGCAGCCCGUUUAGCCGCGUGGUGCAUCUGUAUCG CAACGGCAAA
GLP-1	GACUGGACCUGGAUCCUCUUCUUGGUGGCAGCAGCCACAAGAGUCCACUCCCAU GCUGAAGGGACCUUUACCAGUGAUGUAAGUUCUUAUUUGGAAGGCCAAGCUGCC AAGGAAUUCAUUGCUUGGCUGGUGAAAGGCCGAGGA
VSVG-EPM-EABR-Intein-MCP	 ATGAAGTGCCTTTTGTACTTAGCCTTTTTATTCATTGGGGTGAATTGCAAGTTCACCAT AGTTTTTCCACACAACCAAAAAGGAAACTGGAAAAATGTTCCTTCTAATTACCATTATT GCCCGTCAAGCTCAGATTTAAATTGGCATAATGACTTAATAGGCACAGCCTTACAAGT CAAAATGCCCAAGAGTCACAAGGCTATTCAAGCAGACGGTTGGATGTGTCATGCTTC CAAATGGGTCACTACTTGTGATTTCCGCTGGTATGGACCGAAGTATATAACACATTCC ATCCGATCCTTCACTCCATCTGTAGAACAATGCAAGGAAAGCATTGAACAAACGAAAC AAGGAACTTGGCTGAATCCAGGCTTCCCTCCTCAAAGTTGTGGATATGCAACTGTGA CGGATGCCGAAGCAGTGATTGTCCAGGTGACTCCTCACCATGTGCTGGTTGATGAAT ACACAGGAGAATGGGTTGATTCACAGTTCATCAACGGAAAATGCAGCAATTACATATG CCCCACTGTCCATAACTCTACAACCTGGCATTCTGACTATAAGGTCAAAGGGCTATGT GATTCTAACCTCATTTCCATGGACATCACCTTCTTCTCAGAGGACGGAGAGCTATCAT CCCTGGGAAAGGAGGGCACAGGGTTCAGAAGTAACTACTTTGCTTATGAAACTGGAG GCAAGGCCTGCAAAATGCAATACTGCAAGCATTGGGGAGTCAGACTCCCATCAGGT GTCTGGTTCGAGATGGCTGATAAGGATCTCTTTGCTGCAGCCAGATTCCCTGAATGC CCAGAAGGGTCAAGTATCTCTGCTCCATCTCAGACCTCAGTGGATGTAAGTCTAATTC AGGACGTTGAGAGGATCTTGGATTATTCCCTCTGCCAAGAAACCTGGAGCAAAATCA GAGCGGGTCTTCCAATCTCTCCAGTGGATCTCAGCTATCTTGCTCCTAAAAACCCAG GAACCGGTCCTGCTTTCACCATAATCAATGGTACCCTAAAATACTTTGAGACCAGATA CATCAGAGTCGATATTGCTGCTCCAATCCTCTCAAGAATGGTCGGAATGATCAGTGGA ACTACCACAGAAAGGGAACTGTGGGATGACTGGGCACCATATGAAGACGTGGAAATT GGACCCAATGGAGTTCTGAGGACCAGTTCAGGATATAAGTTTCCTTTATACATGATTG GACATGGTATGTTGGACTCCGATCTTCATCTTAGCTCAAAGGCTCAGGTGTTCGAACA

	TCCTCACATTCAAGACGCTGCTTCGCAACTTCCTGATGATGAGAGTTTATTTTTTGGT GATACTGGGCTATCCAAAAATCCAATCGAGCTTGTAGAAGGTTGGTTCAGTAGTTGGA AAAGCTCTATTGCCTCTTTTTTCTTTATCATAGGGTTAATCATTGGACTATTCTTGGTTC TCCGAGTTGGTATCCATCTTTGCATTAAATTAAAGCACACCAAGAAAAGACAGATTTAT ACAGACATAGAGATGAACCGACTTGGAAAGTCTAGAGCCCTGCCAGGAAATCCCGAC CATAGAGAAATGGGCGAAACGCTGCCGGAAGAGGTGGGAGAGTACCGGCAGCCTT CTGGTGGATCCGTTCCCGTGAGCCCTGGACCACCGTCAGGGTTGGAGCCCACCAG CAGCTCCCCATATGGTGGTGGTTCTTTCAATTCCTCCATCAACAACATCCACGAAATG GAAATTCAGCTGAAGGATGCGCTCGAGAAAAACCAACAGTGGTTGGTGTATGATCAG CAGAGAGAAGTATACGTCAAGGGTTTGCTGGCTAAGATCTTCGAGCTCGAGAAAAAG ACCGAAACAGCAGCACATAGCCTTCCCGGTGGTGGTGGTTCTGGTGGTGGTGGTTC TGAATTCGCACTTGCAGAGGGCACAAGAATCTTTGATCCCGTCACCGGAACTACACA CAGGATCGAGGATGTAGTCGGCGGCCGCAAACCAATCCATGTCGTGGCCGCAGCAA AGGACGGAACCCTTCACGCCAGACCGGTGGTGTCCTGGTTCGATCAGGGAACCAG GGACGTCATTGGCCTCAGGATCGCAGGAGGGGCCATCTTGTGGGCCACACCGGAC CACAAGGTTCTGACTGAGTATGGCTGGAGAGCCGCCGGAGAACTGCGAAAGGGCG ATAGAGTAGCTCAGCCCCGGCGGTTTGATGGTTTCGGCGACTCTGCTCCAATACCTG CCGTTCAGGCGCTCGCTGACGCCCTGGACGACAAGTTTCTCCACGACATGCTCGCC GAGGAGTTGAGATACTCCGTGATCCGCGAGGTGCTGCCAACAAGGAGAGCCCGCA CGTTCGGGCTGGAAGTCGAAGAGTTGCACACCCTCGTAGCTGAGGGCGTGGTCGT CCACAATTGTGAATTCGCCTCTAATTTTACTCAATTTGTGCTTGTCGATAATGGGGGGA CGGGAGATGTGACCGTTGCCCCTAGCAATTTCGCAAATGGCGTTGCAGAATGGATCT CTAGCAACAGCAGAAGCCAAGCGTACAAAGTAACGTGTTCCGTTCGCCAAAGCTCC GCCCAAAAACGGAAGTATACAATAAAGGTTGAGGTGCCGAAAGTAGCCACTCAAACA GTTGGTGGGGTAGAATTGCCCGTAGCGGCATGGCGGTCATATCTCAATATGGAACTC ACTATCCCAATCTTCGCCACGAATAGCGATTGTGAGCTGATAGTTAAGGCTATGCAAG GTCTTCTCAAAGATGGAAACCCTATTCCATCTGCTATCGCCGCCAACAGCGGGATATA CTAG
EGFP-6×MS2	5’ EGFP 6xMS2 3’ ATGGTGAGCAAGGGCGAGGAGCTGTTCACCGGGGTGGTGCCCATCCTGGTCGAGC TGGACGGCGACGTAAACGGCCACAAGTTCAGCGTGTCCGGCGAGGGCGAGGGCGA TGCCACCTACGGCAAGCTGACCCTGAAGTTCATCTGCACCACCGGCAAGCTGCCCG TGCCCTGGCCCACCCTCGTGACCACCCTGACCTACGGCGTGCAGTGCTTCAGCCG CTACCCCGACCACATGAAGCAGCACGACTTCTTCAAGTCCGCCATGCCCGAAGGCT ACGTCCAGGAGCGCACCATCTTCTTCAAGGACGACGGCAACTACAAGACCCGCGCC GAGGTGAAGTTCGAGGGCGACACCCTGGTGAACCGCATCGAGCTGAAGGGCATCG ACTTCAAGGAGGACGGCAACATCCTGGGGCACAAGCTGGAGTACAACTACAACAGC CACAACGTCTATATCATGGCCGACAAGCAGAAGAACGGCATCAAGGTGAACTTCAAG ATCCGCCACAACATCGAGGACGGCAGCGTGCAGCTCGCCGACCACTACCAGCAGAA CACCCCCATCGGCGACGGCCCCGTGCTGCTGCCCGACAACCACTACCTGAGCACC CAGTCCGCCCTGAGCAAAGACCCCAACGAGAAGCGCGATCACATGGTCCTGCTGGA GTTCGTGACCGCCGCCGGGATCACTCTCGGCATGGACGAGCTGTACAAGTAGCCGG CGGCCGCCTTGCAGCACATACATGAGGATCACCCATGTCTGCAGGTCGACTCTAGAA AACATGAGGATCACCCATGTCTGCAGTATTCCCGGGTTCATTAGATCCTAAGGTACCT AATTGCCTAGAAAACATGAGGATCACCCATGTCTGCAGGTCGACTCTAGAAAACATGA GGATCACCCATGTCTGCAGTATTCCCGGGTTCATTAGATCCTAAGGTACCTAATTGCC TAGAAAACATGAGGATCACCCATGTCTGCAGGTCGACTCTAGAAAACATGAGGATCA CCCATGTGTGCAC
Twister-5’-ribozyme-BBV-3×Flag-P2A-2×MS2-Twister-3’-ribozyme	5’ Twister BBV 3×Flag P2A 2×MS2 Twister 3’ GCCATCAGTCGCCGGTCCCAAGCCCGGATAAAATGGGAGGGGGCGGGAAACCGCC TAACCATGCCGACTGATGGCAGGTTTTCGAAACAAATAAAACAGAAAAGCGAACCTAA ACACGCCACCATGGACTACAAGGATCATGACGGAGACTATAAGGATCACGATATTGAT TACAAAGATGACGACGACAAAGCTACTAACTTCAGCCTGCTGAAGCAGGCTGGAGAC GTGGAGGAGAACCCTGGACCTGCGGCCGCCTTGCAGCACATACATGAGGATCACCC ATGTCTGCAGGTCGACTCTAGAAAACATGAGGATCACCCATGTCTGCAGTATTCCCG GGTTCATCTGCCATCAGTCGGCGTGGACTGTAGAACACTGCCAATGCCGGTCCCAA GCCCGGATAAAAGTGGAGGGTACAGTCCACGCTTTTTTTTTT

Supplementary Table 4 | Primer sequences.

Primer name	Sequence (5'-3')
IFN-β1-F	CTTGGATTCCTACAAAGAAGCAGC
IFN-β1-R	TCCTCCTTCTGGAACTGCTGCA
TNF-F	CTCTAATCAGCCCTCTGGC
TNF-R	GAGGGTTTGCTACAACATGG
DDX58-F	CACCTCAGTTGCTGATGAAGGC
DDX58-R	GTCAGAAGGAAGCACTTGCTACC
IL-6-F	GCCCTGAGAAAGGAGACAT
IL-6-R	CTGTTCTGGAGGTACTCTAGGTAT
CCL5-F	CCTGCTGCTTTGCCTACATTGC
CCL5-R	ACACACTTGGCGGTTCTTTCGG
CXCL10-F	CTGACTCTAAGTGGCATT
CXCL10-R	TGATGGCCTTCGATTCTG
CCL2-F	AGAATCACCAGCAGCAAGTGTCC
CCL2-R	TCCTGAACCCACTTCTGCTTGG
IL12A-F	TGCTCCAGAAGGCCAGAC
IL12A-R	TTCATAAATACTACTAAGGCACAGG
IFNA1-F	AGAAGGCTCCAGCCATCTCTGT
IFNA1-R	TGCTGGTAGAGTTCGGTGCAGA
IL-1B-F	ACAGATGAAGTGCTCCTTCCA
IL-1B-R	GTCGGAGATTCGTAGCTGGAT

Supplementary Table 5 | List of antibodies used in this study.

Name	Fluorescence	Dilution F	Clone	Company	Catalog
Fixable Viability Dye		1-1000		BioLegend	423102
CD45	APC	1-400	I3/2.3	BioLegend	147708
CD45	PerCP	1-400	30-F11	BioLegend	103130
CD3	PE-Cy5	1-400	17A2	BioLegend	100274
NK1.1	AF700	1-200	S17016D	BioLegend	156512
CD11b	PerCP	1-250	M1/70	BioLegend	101229
F4/80	BV421	1-250	BM8	BioLegend	123132
H-2K^{b} bound to SIINFEKL	PE	1-200	25-D1.16	BioLegend	141603
CD11c	BV650	1-250	N418	BioLegend	117339
CD3	PE-Cy5	1-400	17A2	BioLegend	100274
CD4	APC-Cy7	1-400	GK1.5	BioLegend	100414
CD4	PE-Cy7	1-400	RM4-4	BioLegend	116016
Foxp3	Pacific blue	1-250	MF-14	BioLegend	126410
PD-1	BV711	1-150	29F.1A12	BioLegend	135231
CD8	APC-Cy7	1-400	53-6.7	BioLegend	100713
CD69	FITC	1-250	H1.2F3	BioLegend	104505
IL-17A	BV605	1-200	TC11-18H10.1	BioLegend	506927
IFNγ	PE	1-200	XMG1.2	BioLegend	505808
Anti-DYKDDDDK Tag	PE/Dazzle 594	1-200	L5	BioLegend	637330
Anti Flag M2 Mouse Monoclonal Antibody		1-4000		Merck	F1804
Anti β-Tubulin Mouse Monoclonal Antibody		1-4000	3G7	CoWin Biosciences	CW0098M
Poxidase AffiniPure Goat Anti-Mouse IgG (H+L) secondary antibody		1-2000		Jackson ImmunoResearch	115035003